DOI: 10.1002/adsc.201300408

Arene C–H Bond Functionalization Coupling with Cyclization of Allenes

Rong Zeng,[a] Juntao Ye,[b] Chunling Fu,[a] and Shengming Ma[a,b,*]

[a] Laboratory of Molecular Recognition and Synthesis, Department of Chemistry, Zhejiang University, Zhejiang, Hangzhou 310027, People's Republic of China
[b] State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, People's Republic of China
Fax: (+86)-21-6260-9305; e-mail: masm@sioc.ac.cn

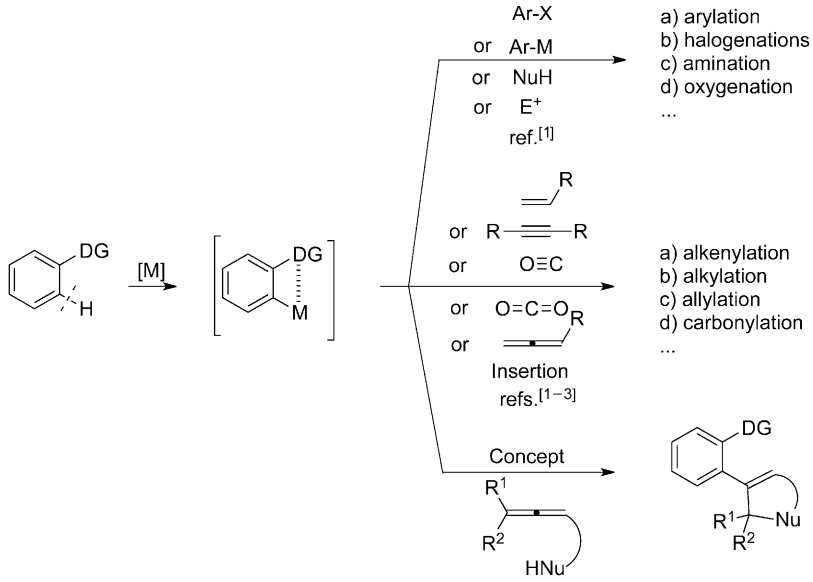
Received: May 9, 2013; Published online: July 12, 2013

Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/adsc.201300408.

Abstract: Under the catalysis of a rhodium(III) complex, the cyclization of 2,3-allenols was most probably initiated with the C–H bond functionalization of *N*-methoxybenzamides using the tetramethylcyclopentadiene anion ($C_5Me_4^-$) as ligand, affording differently substituted 2,5-dihydrofuran derivatives. Highly efficient axial-to-central chirality transfer has been observed.

Keywords: 2,3-allenols; C–H activation; 2,5-dihydrofurans; coupling–cyclization; rhodium catalysts

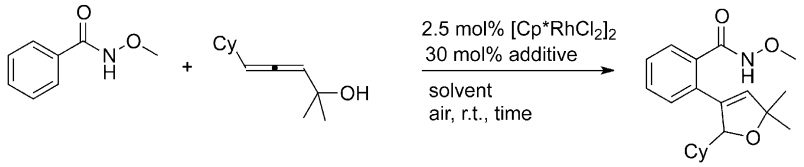
The transition metal-catalyzed C–H functionalization of arenes has been extensively explored to form C–C bond or C–X bond(s) *via* reactions such as arylation, alkenylation, alkylation, halogenations (I, Br, Cl, F), amination, oxygenation, etc. (Scheme 1).[1–3] On the other hand, transition metal-catalyzed cyclization reactions of allenes with a nucleophilic functionality in the presence or absence of organic halides have been proven to be a powerful and efficient tool for the construction of cyclic compounds.[4–5] We envisioned a coupling–cyclization of functionalized allenes involving the functionalization of a C–H bond in arenes (Ar–H).[3] The challenge is the matching of the reactivity of the organometallic intermediates formed from the C–H bond metalation with the cyclization of allenes. Herein, we wish to report the realization of such a concept: Rh(III)-catalyzed coupling–cyclization reactions of allenols with *N*-methoxybenzamides with the oxygen in air as the oxidant,[6,7] providing 2,5-dihydrofuran derivatives, which play important roles in organic chemistry, not only as key structural



Scheme 1. C–H functionalization/coupling reactions.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Table 1. Optimization of the cyclization reaction conditions of allenol **2a** in the presence of *N*-methoxybenzamide **1a**[a]



1a 1.2 equiv. **2a** **3aa**

Entry	Solvent	Additive	Time [h]	NMR Yield[b]	
				2a	**3aa**
1	$MeOH/H_2O$[c]	CsOAc	22	–	57
2	MeOH	CsOAc	22	–	54
3	toluene	CsOAc	22	12	37
4	DMF	CsOAc	40	88	14
5	*t*-BuOH	CsOAc	40	38	34
6	$MeOH/H_2O$[c]	$LiOAc{\cdot}2H_2O$	23	–	58
7	$MeOH/H_2O$[c]	$Cu(OAc)_2{\cdot}H_2O$[d]	24	5	53
8	$MeOH/H_2O$[c]	NaOAc	23	–	62
9	$MeOH/H_2O$[c]	EtCOONa	26	–	46
10	$MeOH/H_2O$[c]	NaOAc[e]	43	63	–
11	$MeOH/H_2O$[c]	NaOAc[f]	43	93	–

[a] The reaction was conducted with **1a** (0.36 mmol), **2a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (0.0075 mmol), CsOAc (0.09 mmol), MeOH (2 mL), and H_2O (0.1 mL).
[b] A small amount of an unidentified product was detected in the reaction in all cases.
[c] $MeOH/H_2O = 20/1$.
[d] 15 mol% of $Cu(OAc)_2{\cdot}H_2O$ was used.
[e] $[Cp^*IrCl_2]_2$ was used instead of $[Cp^*RhCl_2]_2$ as the catalyst.
[f] 5 mol% of $[Cp^*RuCl_2]$ was used instead of $[Cp^*RhCl_2]_2$ as the catalyst.

units in many natural products and pharmaceuticals but also as useful building blocks in synthetic chemistry.[8]

As a first try, the Rh-catalyzed coupling–cyclization reaction of 1.0 equiv. of 5-cyclohexyl-2-methylpenta-3,4-dien-2-ol **2a** with 1.2 equiv. of *N*-methoxybenzamide **1a** catalyzed by 2.5 mol% of $[Cp^*RhCl_2]_2$ with 30 mol% of CsOAc was conducted at room temperature in $MeOH/H_2O$ (20/1). Gratifyingly, the reaction did afford the expected coupling–cyclization product **3aa** in 57% NMR yield (entry 1, Table 1), together with a small amount of an unidentified product. Then the solvent effect was explored: MeOH gave a lower yield (54%) with no **2a** being recovered (entry 2, Table 1); toluene and *t*-BuOH only afforded 37% and 34% yields of product **3aa** together with 12% and 38% recovery of allenol **2a**, respectively (entries 3 and 5, Table 1); DMF led to the worst result: 14% of **3aa** together with 88% recovery of **2a** after stirring for 40 h (entry 4, Table 1). Different inorganic acetates were then screened: $LiOAc{\cdot}2H_2O$ gave the similar result yielding **3aa** in 58% yield (entry 6, Table 1), while $Cu(OAc)_2$,[5,6] which also could be an oxidant, gave a lower yield (53%) (entry 7, Table 1); to our delight, the readily available and inexpensive NaOAc gave a higher yield of **3aa** (62%) (entry 8, Table 1); EtCOONa led to a lower yield (46%) (entry 9, Table 1). In the presence of NaOAc, the reactions catalyzed by $[Cp^*IrCl_2]_2$ or $[Cp^*RuCl_2]_2$ proceeded with a much higher recovery (entries 10 and 11, Table 1).

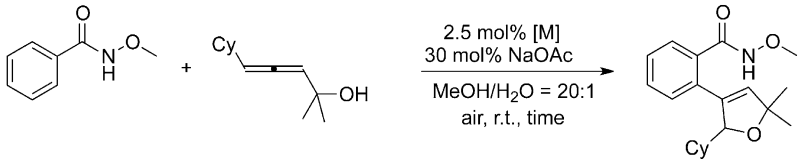
Furthermore, it is interesting to observe that $RhCl_3$ failed to afford any expected product (entry 2, Table 2); $[CpMe_4RhCl_2]_2$ was then used to catalyze the reaction with the purpose of reducing the steric hindrance of the applied ligand, which may favor the C–H bond functionalization and oxymetalation. Luckily, the yield of **3aa** was improved to 66% (entry 3, Table 2), and could be further increased to 70% by adding 10 mol% of additional tetramethylcyclopentadiene ($C_5H_2Me_4$) (entry 4, Table 2). This steric effect of the ligand was further confirmed by using the more sterically hindered ligand tetraphenylcyclopentadiene ($C_5H_2Ph_4$) with 50% yield of **3aa** (entry 5, Table 2). Increasing the amount of $C_5H_2Me_4$ to 20 mol% afforded a lower yield with a trace recovery of **2a** (entry 6, Table 2). Surprisingly, the yield could be improved to 72% by decreasing the loading of **1a** to 1.0 equiv. (entry 7, Table 2). Thus, we defined the coupling–cyclization reaction of allenols and 1.0 equiv. of *N*-methoxybenzamides in $MeOH/H_2O$ at 25 °C in air in the presence of 2.5 mol% of $[CpMe_4RhCl_2]_2$, 10 mol% of $C_5H_2Me_4$, and 30 mol% of NaOAc as the standard reaction conditions for further investigation.

With the optimized conditions in hand, the scope of the cyclization–coupling reactions was explored with

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. The ligand effect on the cyclization of **2a** in the presence of **1a**.

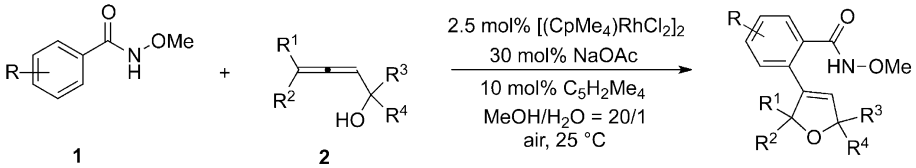


1a 1.2 equiv. **2a** **3aa**

Entry	M	Additive (mol%)	NMR Yield of **3aa**
1	$[Cp^*RhCl_2]_2$	–	62
2	$RhCl_3{\cdot}xH_2O$[a]	**C_5HMe_5** (10)[b]	–[c]
3	**$[CpMe_4RhCl_2]_2$**	–	66
4	**$[CpMe_4RhCl_2]_2$**	**$C_5H_2Me_4$** (10)	70
5	**$[CpMe_4RhCl_2]_2$**	$C_5H_2Ph_4$ (10)	50[d]
6	**$[CpMe_4RhCl_2]_2$**	$C_5H_2Me_4$ (20)	66[e]
7[f]	**$[CpMe_4RhCl_2]_2$**	$C_5H_2Me_4$ (10)	72

[a] Rh% = 40%.
[b] 20 mol% of $NaHCO_3$ was used.
[c] 97% of **2a** was recovered.
[d] 21% of **2a** was recovered.
[e] 11% of **2a** was recovered.
[f] 1.0 equiv. of **1a** was used.

Table 3. Rh(III)-catalyzed cyclization/oxidative coupling of *N*-methoxybenzamides and 2,3-allenols.[a]



1 1.0 equiv. **2** 1.0 equiv. **3**

Entry	**1** R	**2** $R^1/R^2/R^3/R^4$	Time [h]	Isolated yield of **3** [%]
1	H (**1a**)	Cy/H/Me/Me (**2a**)	30	66 (**3aa**)
2	H (**1a**)	Cy/H/-$(CH_2)_5$- (**2b**)	28	72 (**3ab**)
3	H (**1a**)	*i*-Pr/H/-$(CH_2)_5$- (**2c**)	23	70 (**3ac**)
4	H (**1a**)	*i*-Pr/H/Me/*i*-Pr (**2d**)	24	74 (**3ad**)[b]
5[c]	H (**1a**)	*i*-Pr/H/Ph/Ph (**2e**)	33	78 (**3ae**)
6	H (**1a**)	-$(CH_2)_5$-/-$(CH_2)_5$- (**2f**)	18	67 (**3af**)[d]
7	H (**1a**)	-$(CH_2)_5$-/Me/Ph (**2g**)	17	43 (**3ag**)[d]
8	4-Br (**1b**)	**2b**	26	63 (**3bb**)
9	4-MeO (**1c**)	**2b**	32	70 (**3cb**)
10	4-COOMe (**1h**)	**2b**	13	68 (**3hb**)
11	3-NO_2 (**1d**)	**2b**	20	51 (**3db**)
12	3-CF_3 (**1e**)	**2b**	24	83 (**3eb**)
13	2-Cl-4-Br (**1f**)	**2b**	20	59 (**3fb**)
14[e]	H (**1a**)	**2b**	33	69 (**3ab**)

[a] The reaction was conducted with **1** (1.0 mmol), **2** (1.0 mmol), $[CpMe_4RhCl_2]_2$ (0.025 mmol), NaOAc (0.3 mmol), MeOH (6 mL), H_2O (0.3 mL), and monitored by TLC.
[b] *dr* = 47/53.
[c] 1.16 equiv. of **2e** were used.
[d] The reaction was conducted at 10 °C.
[e] The reaction was conducted on a 10.0-mmol scale (2.20 g) of **2b**.

a variety of differently substituted *N*-methoxybenzamides (**1a–1f** and **1h**) and allenols (**2a–2g**), and the results are outlined in Table 3. Differently 4-substituted 2,3-allenols could be applied: the R^3, R^4 groups could be alkyl (**2a–2d**) or aryl groups (**2e** and **2g**). 4,4-Disubstituted allenols (**2f–2g**) could also be applied in

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

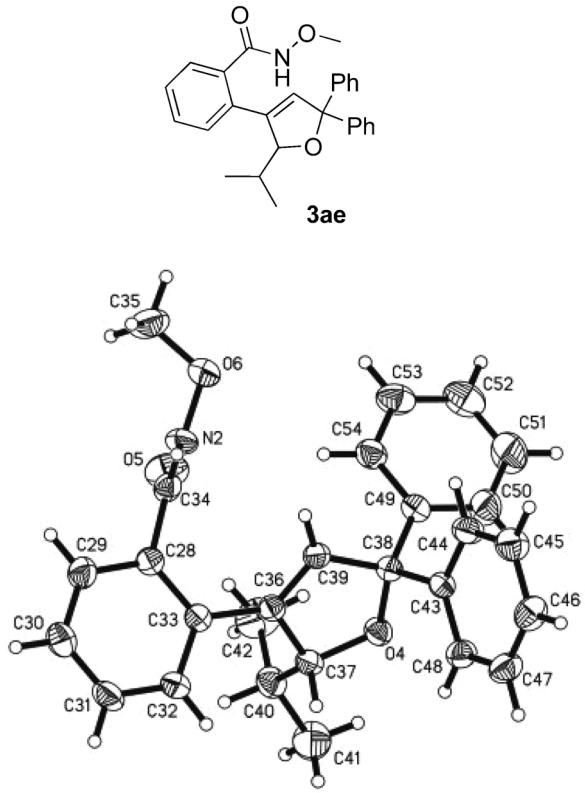


Figure 1. ORTEP representation of **3ae**.

the reaction by decreasing the temperature to 10 °C, and the corresponding coupling–cyclization products may be obtained in moderate yields. Using allenol **2b**, differently substituted *N*-methoxybenzamides **1** were also applied: either electron-donating substituents such as 4-OMe (**1c**), or electron-withdrawing groups, such as 4-Br (**1b**), 4-COOMe (**1h**), 3-NO_2 (**1d**), 3-CF_3 (**1e**), and 2-Cl-4-Br groups (**1f**) are accommodated. Finally, the reaction can be easily conducted on a 10-mmol scale of the substrate **2b** in a similar yield (entry 14, Table 3). The structures of the compounds were further conformed by the X-ray single crystal diffraction study of **3ae** (Figure 1).[9]

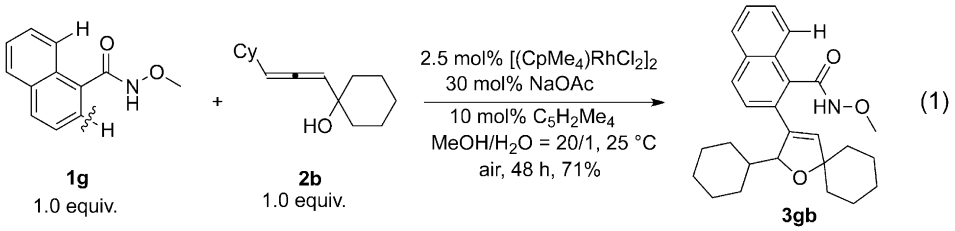
In addition, when *N*-methoxy-1-naphthylformamide **1g** was applied, the 2-position C–H bond reacted exclusively to afford the corresponding product **3gb** in 71% yield [Eq. (1)], no reaction occurred at the 8-position C–H bond of the naphthyl ring.[10]

Four possible pathways have been proposed for this coupling–cyclization reaction as summarized in Scheme 2: (i) C–H rhodation of **1a** provides the intermediate **M1**,[3e,f,11] which is followed by the coordination of the allene moiety in **2b** with Rh and cyclic oxyrhodation to afford **M2**. Reductive elimination would yield the product **3ab** (path **A**, Scheme 2); (ii) carbometallation of allenol **2b** with **M1** would afford the π-allyl rhodium intermediate **M3**,[13–14] which would yield **3ab** by intramolecular nucleophilic substitution with the hydroxy group (path **B**, Scheme 2); (iii) cyclic oxyrhodation of **2b** with Rh(III) would afford the 2,5-dihydrofuryl rhodium intermediate **M4**, which may be followed by aryl rhodation with **1a** to afford **M2**. Reductive elimination would yield **3ab** (path **C**, Scheme 2); (iv) protonolysis of **M4** would afford **4b**, which would couple with **1a** to afford **M2** and subsequently **3ab** (path **D**, Scheme 2).

To further probe the reaction mechanism, firstly, the KIE (k_H/k_D) was determined to be 4:1 or 5:1, respectively, by running the reaction shown in Eq. (2) or Eq. (3), indicating that the C–H bond cleavage process should be a crucial step.

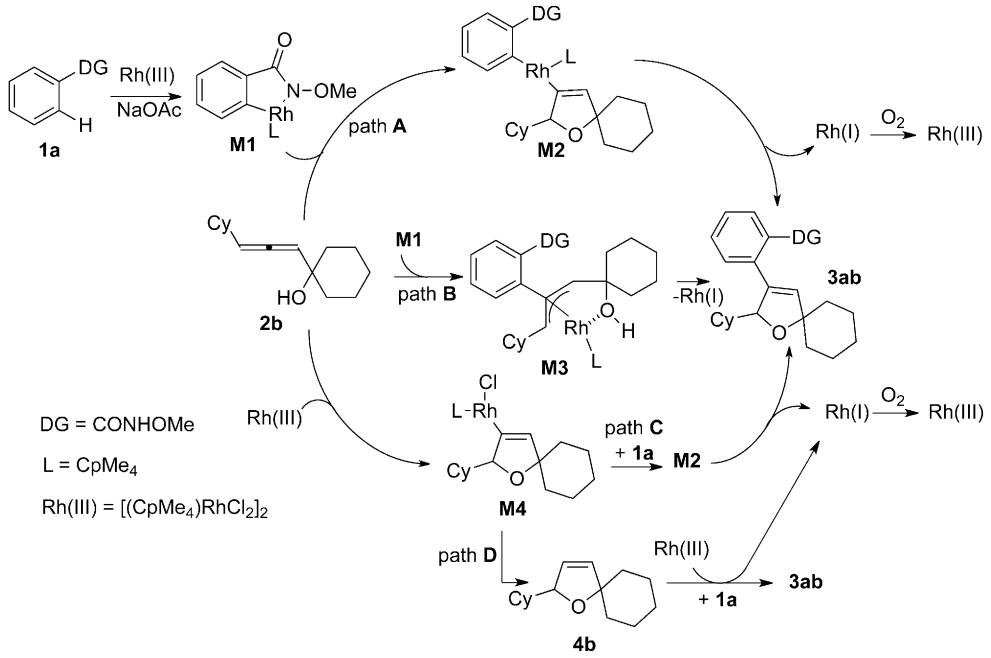
Moreover, the cyclization of **2b** was carried out under the standard conditions in the absence of *N*-methoxybenzamide **1a** (upper reaction in Scheme 3). Interestingly, the reaction only afforded 10% of cycloisomerization product **4b** together with 76% recovery of **2b** even after 25 h; the yield of **4b** could be improved to 88% only when 50 mol% of the catalyst was used. These results indicated that the formation of intermediate **M4** in Scheme 2 was a rather slow process in the reaction, indicating that at least path **C** may not be the predominant pathway. It was also observed that the reaction of the cycloisomerization product **4b** with **1a** failed to afford the product **3ab**, excluding the possibility of path **D** (lower reaction in Scheme 3).

Furthermore, the optically active allenols *R*-**2b** and *S*-**2b** were synthesized[12] and submitted to the reactions with *N*-methoxybenzamides **1a**–**1c** under the standard conditions. Gratifyingly, the efficient axial-to-central chirality transfer from *R*-**2b** and *S*-**2b** to the corresponding optically active 2,5-dihydroduran derivatives was observed (Table 4). These results may mostly exclude the path **B**, in which the σ-π-σ rearrangement of π-allyl rhodium intermediate **M3** would lead to racemization.[4,13–16] In conclusion, although we

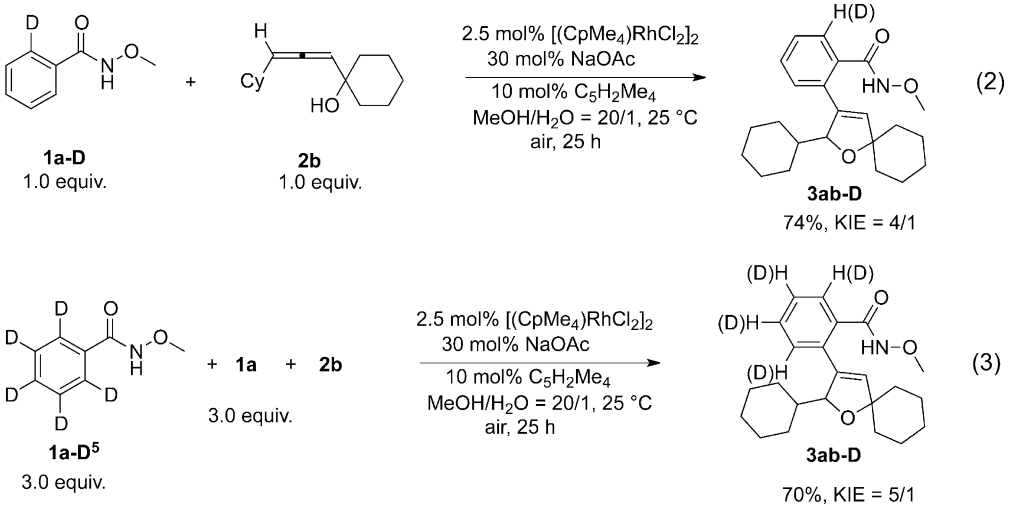


 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

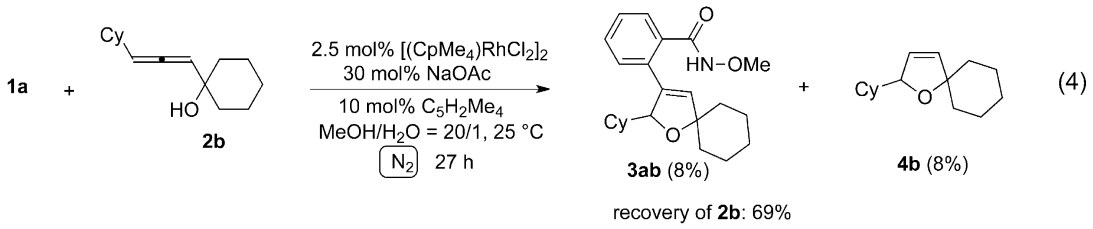


Scheme 2. Four possible pathways.



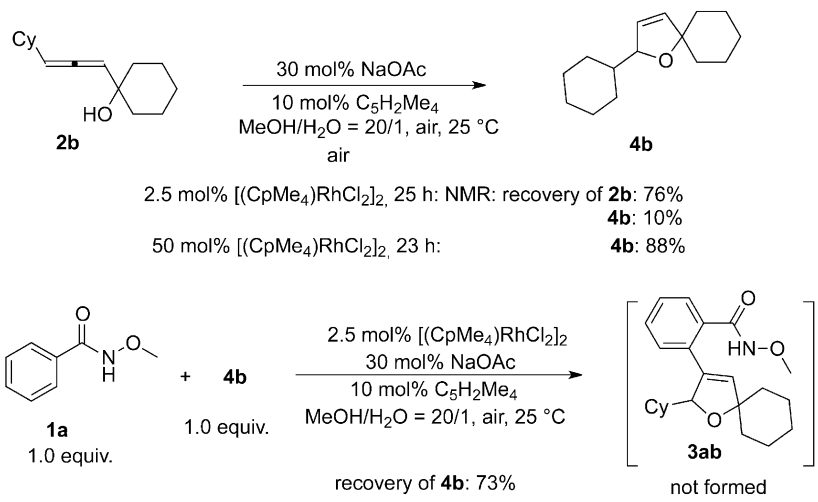
may not be able to exclusively rule out paths **B** and **C**, the reaction should proceed mostly probably *via* path **A** in Scheme 2.

Finally, when the reaction was conducted under a nitrogen atmosphere, only 8% of product was observed together with 69% of allenol recovery [Eq. (4)], indicating that the oxygen in the air must be an efficient oxidant for the *in-situ* generated Rh(I) to regenerate Rh(III) furnishing the catalytic cycle for this transformation.[7]



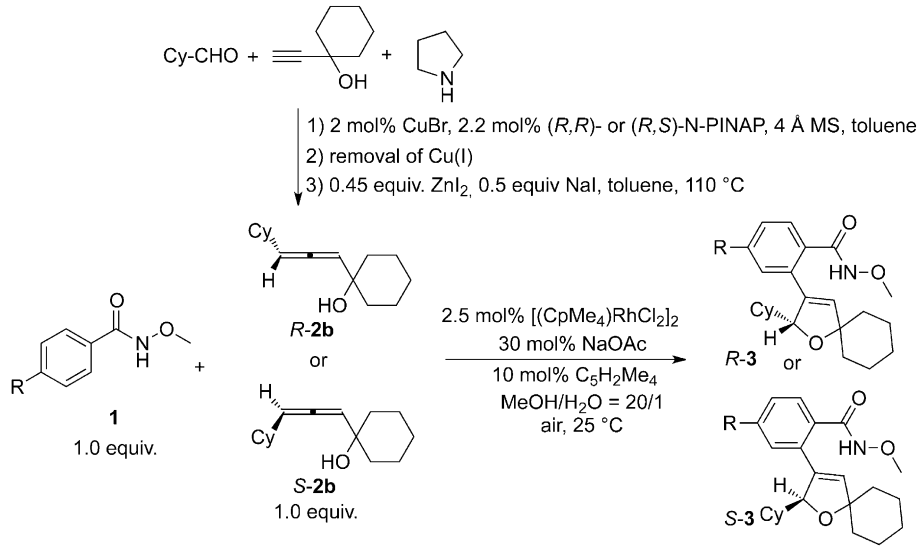
 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Scheme 3. Mechanistic studies.

Table 4. The axial-to-central chirality transfer reactions of 2,3-allenols *R*-**2b** and *S*-**2b**.



Entry	R	*ee* value of **2b**	Time [h]	Yield [%] (**3**) (*ee*)
1	H (**1a**)	95% (*R*-**2b**)	26	68 (*R*-**3ab**) (94%)
2	Br (**1b**)	95% (*R*-**2b**)	19	68 (*R*-**3bb**) (92%)
3	OMe (**1c**)	95% (*R*-**2b**)	34	74 (*R*-**3cb**) (94%)
4	H (**1a**)	95% (*S*-**2b**)	26.5	76 (*S*-**3ab**) (93%)

In summary, we have developed a rhodium-catalyzed coupling–cyclization reaction of allenols with *N*-methoxybenzamides using the tetramethylcyclopentadienyl group as ligand. Optically active 2,5-dihydrofurans could be synthesized by using axially chiral 2,3-allenols[12] through an efficient axial-to-central chirality transfer process. Mechanistic experiments indicated that the reaction most likely first involves C–H rhodation of arenes, which is followed by cyclic oxyrhodation of allenols and reductive elimination to afford the final coupling–cyclization products. The *in-situ* generated Rh(I) would be reoxidized to Rh(III) with the O_2 in the air to finish the catalytic cycle. Further studies in this area including a further mechanistic study are being carried out in our laboratory.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Advanced Synthesis & Catalysis

Experimental Section

Typical Procedure, Synthesis of 3aa

To a dried Schlenk tube equipped with a Teflon-coated magnetic stirring bar were added **1a** (151.0 mg, 1.0 mmol), $[(CpMe_4)RhCl_2]_2$ (15.0 mg, 0.025 mmol), NaOAc (25.5 mg, 0.3 mmol), tetramethylcyclopentadiene (13.8 mg, 85% purity, 0.1 mmol)/MeOH (2 mL), **2a** (178.9 mg, 1.0 mmol)/MeOH (4 mL), and water (0.3 mL) sequentially at 25 °C. After being stirred for 30 h, the reaction was complete as monitored by TLC, filtration through a short pad of silica gel (eluent: Et_2O) and evaporation afforded the crude product, which was purified by flash column chromatography on silica gel (eluent: 30–60 °C petroleum ether/ethyl acetate = 4/1 to 3.5/1 to 3.2/1) to afford **3aa** as a solid; yield: 215.8 mg (66%); mp 150.7–151.9 °C (hexane/dichloromethane); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.77 (s, 1H, NH), 7.60–7.10 (m, 4H, Ar-H), 5.92 (s, 1H, CH=), 5.11 (s, 1H, CHO), 3.85 (s, 3H, CH_3O), 1.80–0.80 (m, 17H, $CH_2\times5$ and $CH_3\times2$ and CH); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 167.4, 138.6, 135.7, 133.2, 131.8, 130.6, 129.0, 128.4, 127.6, 89.8, 86.7, 64.3, 40.7, 30.4, 28.0, 27.9, 26.6, 26.2, 26.1; IR (neat): ν = 3191, 2971, 2929, 2852, 1644, 1594, 1495, 1373, 1356, 1299, 1275, 1155, 1111, 1056, 1040 cm^{-1}; MS (EI, 70 eV): m/z (%) = 330 (M^+ + 1, 16.98), 328 (M^+–1, 1.76), 199 (100); anal. calcd. for $C_{20}H_{27}NO_3$ (MW = 329): C 72.92, H 8.26, N 4.25; found: C 72.89, H 8.10, N 4.16.

Acknowledgements

*Financial support from National Natural Science Foundation of China (No. 21232006) and National Basic Research Program of China (No. 2009CB825300) is gratefully acknowledged. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University. We thank Mr. Can Zhu in this group for reproducing the results for synthesis of compounds **3ac**, **3ae**, and S-**3ab**.*

References

[1] For recent reviews on C–H functionalization, see: a) D. A. Colby, R. G. Bergman, J. A. Ellman, *Chem. Rev.* **2010**, *110*, 624; b) J. L. Bras, J. Muzart, *Chem. Rev.* **2011**, *111*, 1170; c) C. S. Yeung, V. M. Dong, *Chem. Rev.* **2011**, *111*, 1215; d) L. Ackermann, *Chem. Rev.* **2011**, *111*, 1315; e) T. W. Lyons, M. S. Sanford, *Chem. Rev.* **2010**, *110*, 1147; f) C.-L. Sun, B.-J. Li, Z.-J. Shi, *Chem. Rev.* **2011**, *111*, 1293; g) K. M. Engle, T.-S. Mei, M. Wasa, J.-Q. Yu, *Acc. Chem. Res.* **2012**, *45*, 788; h) S. H. Cho, J. Y. Kim, J. Kwak, S. Chang, *Chem. Soc. Rev.* **2011**, *40*, 5068; i) L. McMurray, F. O'Hara, M. J. Gaunt, *Chem. Soc. Rev.* **2011**, *40*, 1885; j) N. Kuhl, M. N. Hopkinson, J. Wencel-Delord, F. Glorius, *Angew. Chem.* **2012**, *124*, 10382; *Angew. Chem. Int. Ed.* **2012**, *51*, 10236.

[2] For the reports on C-H functionalization with CO_2, see: a) I. I. F. Boogaerts, S. P. Nolan, *J. Am. Chem. Soc.* **2010**, *132*, 8858; b) L. Zhang, J. Cheng, T. Ohishi, Z. Hou, *Angew. Chem.* **2010**, *122*, 8852; *Angew. Chem. Int. Ed.* **2010**, *49*, 8670; c) H. Mizuno, J. Takaya, N. Iwasawa, *J. Am. Chem. Soc.* **2011**, *133*, 1251.

[3] For reports on C–H functionalization involving allenes, see: a) Y. J. Zhang, E. Skucas, M. J. Krische, *Org. Lett.* **2009**, *11*, 4248; b) D. N. Tran, N. Cramer, *Angew. Chem.* **2010**, *122*, 8357; *Angew. Chem. Int. Ed.* **2010**, *49*, 8181; c) D. N. Tran, N. Cramer, *Chimia* **2011**, *65*, 271; d) Y. Kuninobu, P. Yu, K. Takai, *Org. Lett.* **2010**, ##*12*, 4274; e) R. Zeng, C. Fu, S. Ma, *J. Am. Chem. Soc.* **2012**, *134*, 9597; f) B. Ye, N. Cramer, *J. Am. Chem. Soc.* **2013**, *135*, 636; g) H. Wang, F. Glorius, *Angew. Chem.* **2012**, *124*, 7430; *Angew. Chem. Int. Ed.* **2012**, *51*, 7318.

[4] For reviews on cyclization reactions of allenes with a nucleophilic functionality, see: a) R. W. Bates, V. Satcharoen, *Chem. Soc. Rev.* **2002**, *31*, 12; b) S. Ma, *Acc. Chem. Res.* **2003**, *36*, 701; c) S. Ma, in: *Topics in Organometallic Chemistry*, (Ed.: J. Tsuji), Springer-Verlag, Heidelberg, **2005**, pp 183–210.

[5] For reviews on the reactions of allenes, see: a) *Modern Allene Chemistry*, (Eds.: N. Krause, A. S. K. Hashmi), Wiley-VCH, Weinheim, **2004**; b) R. Zimmer, C. U. Dinesh, E. Nandanan, F. A. Khan, *Chem. Rev.* **2000**, *100*, 3067; c) A. Hoffmann-Röder, N. Krause, *Angew. Chem.* **2002**, *114*, 3057; *Angew. Chem. Int. Ed.* **2002**, *41*, 2933; d) L. K. Sydnes, *Chem. Rev.* **2003**, *103*, 1133; e) S. Ma, *Chem. Rev.* **2005**, *105*, 2829; f) S. Ma, *Aldrichimica Acta* **2007**, *40*, 91; g) B. Alcaide, P. Almendros, T. M. del Campo, *Chem. Eur. J.* **2010**, *16*, 5836.

[6] For the reviews on transition metal-catalyzed reactions using oxygen as oxidant, see: a) Z. Shi, C. Zhang, C. Tang, N. Jiao, *Chem. Soc. Rev.* **2012**, *41*, 3381; b) T. Punniyamurthy, S. Velusamy, J. Iqbal, *Chem. Rev.* **2005**, *105*, 2329; c) K. M. Gligorich, M. S. Sigman, *Chem. Commun.* **2009**, 3854; d) S. S. Stahl, *Angew. Chem.* **2004**, *116*, 3480; *Angew. Chem. Int. Ed.* **2004**, *43*, 3400; e) A. N. Campbell, S. S. Stahl, *Acc. Chem. Res.* **2012**, *45*, 851; f) J. Wencel-Delord, T. Dröge, F. Liu, F. Glorius, *Chem. Soc. Rev.* **2011**, *40*, 4740.

[7] For the selected reports on rhodium-catalyzed C–H functionalization using air as the terminal oxidant, see: a) K. Ueura, T. Satoh, M. Miura, *Org. Lett.* **2007**, *9*, 1407; b) D. R. Stuart, P. Alsabeh, M. Kuhn, K. Fagnou, *J. Am. Chem. Soc.* **2010**, *132*, 18326; c) T. K. Hyster, T. Rovis, *Chem. Sci.* **2011**, *2*, 1606; d) N. Schröder, T. Besset, F. Glorius, *Adv. Synth. Catal.* **2012**, *354*, 579; e) L. Zheng, J. Wang, *Chem. Eur. J.* **2012**, *18*, 9699.

[8] a) F. M. Dean, M. V. Sargent, in: *Comprehensive Heterocyclic Chemistry*, (Eds.: A. R. Karitzky, C. W. Rees), Pergamon, Oxford, **1984**, Vol. 4, pp 531–712; b) R. Benassi, in: *Comprehensive Heterocyclic Chemistry* II. (Eds.: A. R. Karitzky, C. W. Rees, E. F. V. Scriven), Pergamon, Oxford, **1996**, Vol. 2, pp 259–436.

[9] **3ae**: $C_{22}H_{27}NO_3$, MW = 413.50, triclinic, space group P–1, final R indices [I > 2σ(I)], $R1$ = 0.0573, $wR2$ = 0.1203; R indices (all data), $R1$ = 0.1220, $wR2$ = 0.1533; a = 9.5620 (7) Å, b = 15.6698 (15) Å, c = 16.8193 (15) Å, α = 67.206 (9)°, β = 89.694 (7)°, γ = 86.030 (7)°, V = 2317.1 (3) Å3, T = 293 (2) K, Z = 4, reflections collected/unique 14716/8480 (R_{int} = 0.0406), number of observations [$>2\sigma(I)$] 4613, parameters: 565. Supplementary crystallographic data have been deposited at the Cam-

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

bridge Crystallographic Data Centre under CCDC 929374. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre *via* www.ccdc.cam.ac.uk/data_request/cif.

[10] For selected reported on C–H functionalization of 8-position of naphthalene, see: a) T. Satoh, Y. Nishinaka, M. Miura, M. Nomura, *Chem. Lett.* **1999**, 615; b) T. Satoh, Y. Kawamura, M. Miura, M. Nomura, *Angew. Chem.* **1997**, *109*, 1820; *Angew. Chem. Int. Ed. Engl.* **1997**, *36*, 1740.

[11] Rh(III)-catalyzed C–H functionalization of *N*-methoxybenzamides **1**, see: a) N. Guimond, C. Gouliaras, K. Fagnou, *J. Am. Chem. Soc.* **2010**, *132*, 6908; b) S. Rakshit, C. Grohmann, T. Besset, F. Glorius, *J. Am. Chem. Soc.* **2011**, *133*, 2350.

[12] a) J. Ye, S. Li, B. Chen, W. Fan, J. Kuang, J. Liu, Y. Liu, B. Miao, B. Wan, Y. Wang, X. Xie, Q. Yu, W. Yuan, S. Ma, *Org. Lett.* **2012**, *14*, 1346; b) J. Ye, W. Fan, S. Ma, *Chem. Eur. J.* **2013**, *19*, 716.

[13] For a review on different insertion reactions of allenes with metallic species, see: T. Bai, S. Ma, G. Jia, *Coord. Chem. Rev.* **2009**, *253*, 423.

[14] For selected reports on insertion of allene affording π-allyl rhodium intermediates, see: a) A. Lumbroso, P. Koschker, N. R. Vautravers, B. Breit, *J. Am. Chem. Soc.* **2011**, *133*, 2386; b) P. Koschker, A. Lumbroso, B. Breit, *J. Am. Chem. Soc.* **2011**, *133*, 20746; c) M. L. Cooke, K. Xu, B. Breit, *Angew. Chem.* **2012**, *124*, 11034; *Angew. Chem. Int. Ed.* **2012**, *51*, 10876.

[15] For a selected report on Rh-catalyzed enantioselective allylic alkylation of propenyl acetate *via* π-allyl rhodium intermediates, see: T. Hayashi, A. Okasa, T. Suzuka, M. Kawatsura, *Org. Lett.* **2003**, *5*, 1713.

[16] For a report on Rh-catalyzed allylic alkylation of 1,3-disubstituted allylic carbonate with partial racemization from 98% *ee* to 90% *ee*, see: P. A. Evans, J. D. Nelson, *J. Am. Chem. Soc.* **1998**, *120*, 5581.

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

16154169, 2013, 10, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201300408 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License